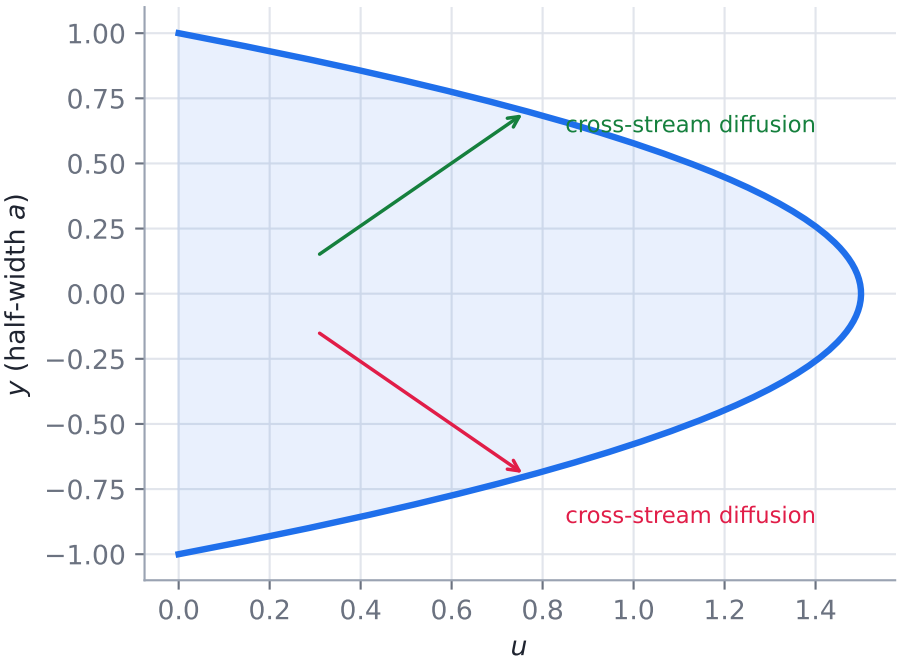
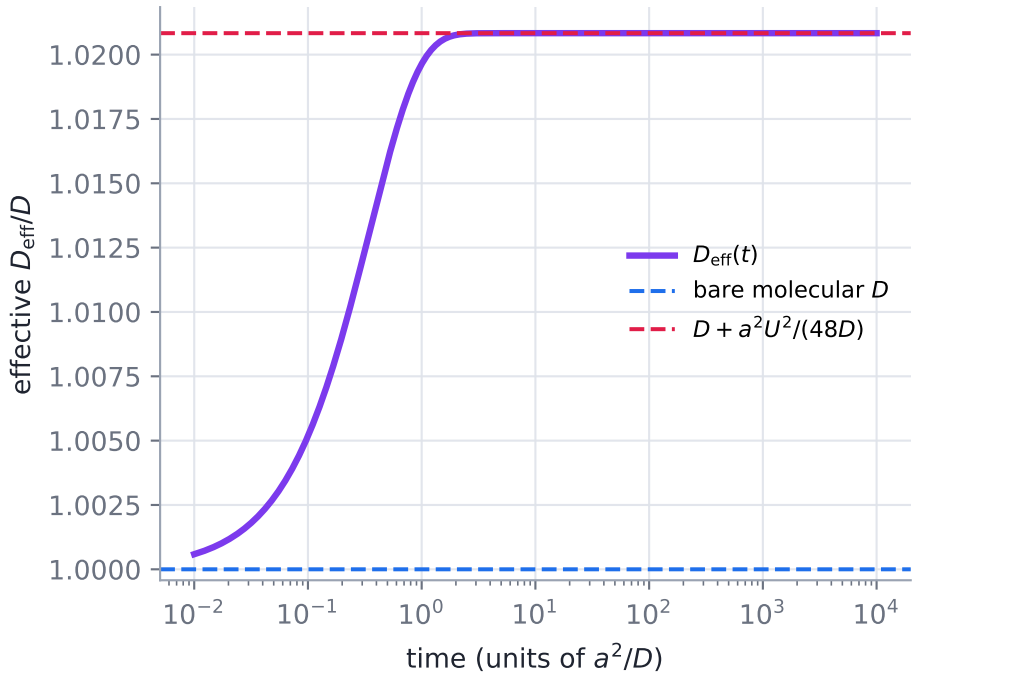

the mechanism (G. I. Taylor, 1953)
fast centre, slow walls, coupled by molecular diffusion
cross-stream diffusion
cross-stream diffusion
y (half-width a)
u
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4


what a tracer test actually measures
early: molecular; late: Fickian again, with a far larger D
effective D_eff/D
time (units of a²/D)
D_eff(t)
bare molecular D
D + a²U²/(48D)
1.0200
1.0175
1.0150
1.0125
1.0100
1.0075
1.0050
1.0025
1.0000
10⁻²
10⁻¹
10⁰
10¹
10²
10³
10⁴